

ISEN/DAEN 427 Lecture 3

Alexander Abuabara

Accuracy Paradox

(Sensitivity and Specificity)



- Engineering often involves **decisions under uncertainty**
- Anomaly detection, quality checks, fraud alerts, ...
- How to balance **false alarms** vs **missed risks**

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{FPR} = 1 - \text{Specificity}$$

T: true, F: false, P: positive, N: negative, R: rate

Confusion Matrix

	Actual Positive	Actual Negative
Predicted Positive	True Positive (TP)	False Positive (FP)
Predicted Negative	False Negative (FN)	True Negative (TN)

Interpretation

- Sensitivity: TP column normalized
- Specificity: TN column normalized

Confusion Matrix

		True class		Measures
		Positive	Negative	
Predicted class	Positive	True positive <i>TP</i>	False positive <i>FP</i>	Positive predictive value (PPV) $\frac{TP}{TP+FP}$
	Negative	False negative <i>FN</i>	True negative <i>TN</i>	Negative predictive value (NPV) $\frac{TN}{FN+TN}$
Measures		Sensitivity $\frac{TP}{TP+FN}$	Specificity $\frac{TN}{FP+TN}$	Accuracy $\frac{TP+TN}{TP+FP+FN+TN}$

- High Sensitivity: catch more issues but more false alarms
- High Specificity: fewer false alarms but risk missing issues
- Balance depends on **cost of errors** and **base rates**

Bayesian Inference

Bayesian inference provides a framework for combining prior knowledge with the likelihood of observed data.

$$P(H|D) = \frac{P(D|H) P(H)}{P(D)}$$

- $P(H|D)$: Posterior probability
- $P(D|H)$: Likelihood (**Sensitivity** when D is a positive test/alert)
- $P(H)$: Prior probability (base rate)
- $P(D)$: Evidence (normalizer)

Odds Form

Posterior Odds = Prior Odds $\times$ Likelihood Ratio,
$$\frac{P(H|D)}{P(\neg H|D)} = \frac{P(H)}{P(\neg H)} \times \frac{P(D|H)}{P(D|\neg H)}$$

Positive Predictive Value (PPV)

Let H = “Issue exists”, D = “System raises an alert”

$$\text{PPV} = P(H|D) = \frac{\text{Sensitivity} \cdot P(H)}{\text{Sensitivity} \cdot P(H) + (1 - \text{Specificity}) \cdot (1 - P(H))}$$

Negative Predictive Value (NPV)

$$\text{NPV} = P(\neg H|\neg D) = \frac{\text{Specificity} \cdot (1 - P(H))}{\text{Specificity} \cdot (1 - P(H)) + (1 - \text{Sensitivity}) \cdot P(H)}$$

Example: Fraud Detection (Base Rate Fallacy)

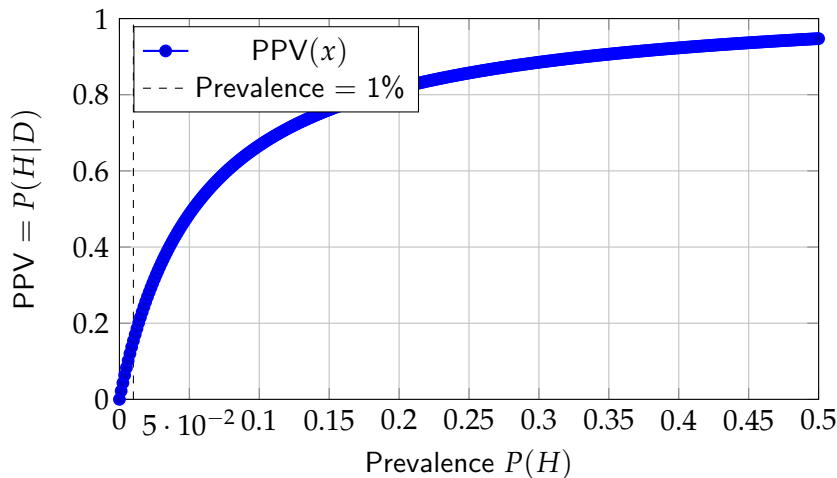
- Base rate: $P(H) = 0.01$
- Sensitivity = 0.90, Specificity = 0.95

$$P(H|D) = \frac{0.9 \cdot 0.01}{0.9 \cdot 0.01 + 0.05 \cdot 0.99} \approx 0.154.$$

Even accurate detectors can yield many false alarms when the true event is rare.

PPV vs. Prevalence (Bayes in Action)

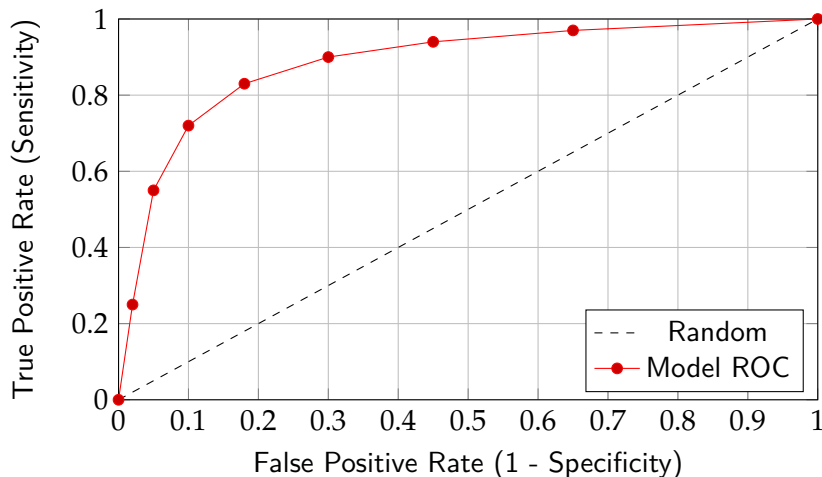
Sensitivity=0.90, Specificity=0.95



PPV rises slowly when base rate is low;
increases faster as prevalence grows.

ROC Curve (Sensitivity vs 1 - Specificity)

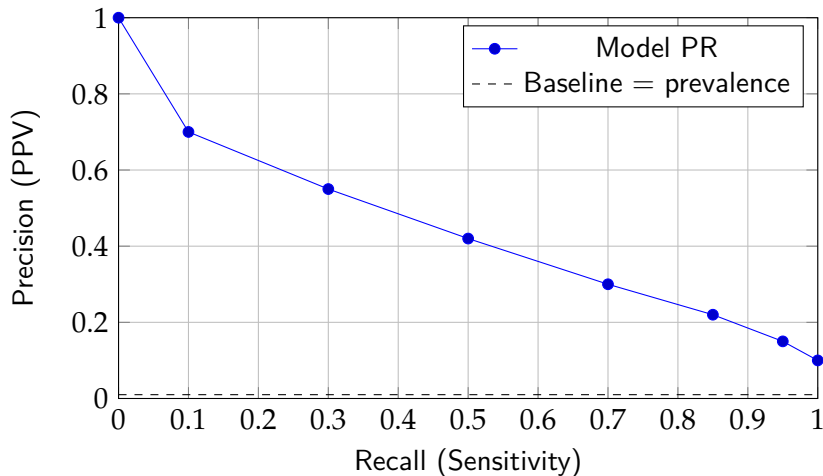
Classifier operating points as threshold varies



Moving the threshold slides you along the ROC curve;
choose based on error costs.

Precision–Recall Curve

Better for rare-event problems



When events are rare,
PR curves reveal usefulness beyond ROC.

Decision-Making Under Uncertainty

- False Positives → wasted resources, alert fatigue
- False Negatives → hidden risks, potential disasters
- Bayesian inference quantifies **true risk given alerts**

Risk Analysis Framework

- 1 Define prior risk (base rate)
- 2 Measure sensitivity & specificity
- 3 Apply Bayes' theorem $\rightarrow$ posterior probability
- 4 Decide threshold based on error costs

- Sensitivity & specificity quantify detection performance
 - Bayes' theorem turns detection metrics into decision probabilities
 - Tune thresholds to match **business risk tolerance** and base rates
-
- For next class,
please think of example(s) of false positives that overwhelm you.
 - In class assignment,
Which is more costly: missed events or false alarms?